



Department of Computer Science and Engineering

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Course Code: CSE 207

Course Title: Data Structures

Lab: 02

Submitted to

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Exercise -1

Code:

```
#include <stdio.h>

void addition(int mat1[50][50], int mat2[50][50], int matn[50][50], int
rows, int columns) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < columns; j++) {
            matn[i][j] = mat1[i][j] + mat2[i][j];
        }
    }

    printf("\nMatrix Addition:\n");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < columns; j++) {
            printf("%d ", matn[i][j]);
        }
        printf("\n");
    }
}
```

```
void subtraction(int mat1[50][50], int mat2[50][50], int matn[50][50],
int rows, int columns) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < columns; j++) {
            matn[i][j] = mat1[i][j] - mat2[i][j];
        }
    }

    printf("\nMatrix Subtraction:\n");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < columns; j++) {
            printf("%d ", matn[i][j]);
        }
        printf("\n");
    }
}
```

```
void multiplication(int mat1[50][50], int mat2[50][50], int mul[50][50],
int rows, int columns) {

    printf("Matrix Multiplication:\n");
```

```
for (int i = 0; i < rows; i++) {  
    for (int j = 0; j < columns; j++) {  
        mul[i][j] = 0;  
        for (int k = 0; k < columns; k++) {  
            mul[i][j] += mat1[i][k] * mat2[k][j];  
        }  
    }  
}
```

```
for (int i = 0; i < rows; i++) {  
    for (int j = 0; j < columns; j++) {  
        printf("%d ", mul[i][j]);  
    }  
    printf("\n");  
}
```

```
int main() {  
    int mat1[50][50], mat2[50][50], matn[50][50], rows, columns, in,  
    mul[50][50];
```

```
printf("Enter the number of rows for the matrix: ");
scanf("%d", &rows);
printf("Enter the number of columns for the matrix: ");
scanf("%d", &columns);
```

```
printf("Enter the first matrix elements:\n");
for (int i = 0; i < rows; i++) {
    for (int j = 0; j < columns; j++) {
        scanf("%d", &mat1[i][j]);
    }
}
```

```
printf("Enter the second matrix elements:\n");
for (int i = 0; i < rows; i++) {
    for (int j = 0; j < columns; j++) {
        scanf("%d", &mat2[i][j]);
    }
}
```

```
printf("\n1 for addition\n");
printf("2 for subtraction\n");
printf("3 for multiplication\n")
```

```
scanf("%d", &in);

if (in == 1) {
    addition(mat1, mat2, matn, rows, columns);
} else if (in == 2) {
    subtraction(mat1, mat2, matn, rows, columns);
} else if (in == 3) {
    multiplication(mat1, mat2, mul, rows, columns);
}

return 0;
}
```

Output:

```
Enter the number of rows for the matrix: 3
Enter the number of columns for the matrix: 3
Enter the first matrix elements:
1
2
3
1
2
3
1
2
3
Enter the second matrix elements:
1
2
3
1
2
3
1
2
3
1 for addition
2 for subtraction
3 for multiplication
3
```

```
Matrix Multiplication:
6 12 18
6 12 18
6 12 18

...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise -2

Code:

```
#include <stdio.h>
```

```
int main() {
```

```
    int mat[50][50], rows, columns, i, j, sum;
```

```
    int choice;
```

```
printf("Enter the number of rows for the matrix: ");  
scanf("%d", &rows);  
printf("Enter the number of columns for the matrix: ");  
scanf("%d", &columns);
```

```
printf("Enter the input for the matrix:\n");  
for (i = 0; i < rows; i++) {  
    for (j = 0; j < columns; j++) {  
        scanf("%d", &mat[i][j]);  
    }  
}
```

```
printf("Choose the operation:\n");  
printf("1. Sum of rows\n");  
printf("2. Sum of columns\n");  
scanf("%d", &choice);
```

```
if (choice == 1) {  
    printf("\nSum of rows:\n");  
    for (i = 0; i < rows; i++) {  
        sum = 0;
```

```
        for (j = 0; j < columns; j++) {  
            sum += mat[i][j];  
        }  
        printf("Row %d: %d\n", i + 1, sum);  
    }  
} else if (choice == 2) {  
    printf("\nSum of columns:\n");  
    for (i = 0; i < columns; i++) {  
        sum = 0;  
        for (j = 0; j < rows; j++) {  
            sum += mat[j][i];  
        }  
        printf("Column %d: %d\n", i + 1, sum);  
    }  
} else {  
    printf("Invalid choice.\n");  
}  
  
return 0;  
}
```

Output:

```
Enter the number of rows for the matrix: 3
Enter the number of columns for the matrix: 3
Enter the input for the matrix:
1
2
3
4
5
6
7
8
9
Choose the operation:
1. Sum of rows
2. Sum of columns
2

Sum of columns:
Column 1: 12
Column 2: 15
Column 3: 18
```

```
1

Sum of rows:
Row 1: 6
Row 2: 15
Row 3: 24

...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise –3

Code:

```
#include <stdio.h>

int main() {
    int row, col;
```

```
printf("Enter the number of rows and columns in the matrix: ");
scanf("%d %d", &row, &col);

if (row <= 0 || col <= 0) {
    printf("Invalid matrix dimensions.\n");
    return 1;
}
int matrix[row][col];
printf("Enter the elements of the matrix:\n");
for (int i = 0; i < row; i++) {
for (int j = 0; j < col; j++) {
    scanf("%d", &matrix[i][j]);
}
}
for (int i = 0; i < row; i++) {
    int max_in_row = matrix[i][0];
    for (int j = 1; j < col; j++) {
        if (matrix[i][j] > max_in_row) {
            max_in_row = matrix[i][j];
        }
    }
    printf("Row %d: %d\n", i + 1, max_in_row);
}
```

```
for (int j = 0; j < col; j++) {  
    int max_in_col = matrix[0][j];  
    for (int i = 1; i < row; i++) {  
        if (matrix[i][j] > max_in_col) {  
            max_in_col = matrix[i][j];  
        }  
    }  
    printf("Column %d: %d\n", j + 1, max_in_col);  
}  
  
return 0;  
}
```

Output:

```
Enter the number of rows and columns in the matrix: 3
3
Enter the elements of the matrix:
1
2
3
4
5
6
7
8
9
Row 1: 3
Row 2: 6
Row 3: 9
Column 1: 7
Column 2: 8
Column 3: 9

...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise – 4

Code:

```
#include <stdio.h>

int main() {
    int mat[10][10], rows, columns, i, j;
    printf("Enter the number of rows and columns for the matrix: ");
    scanf("%d %d", &rows, &columns);

    printf("Enter input for matrix:\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < columns; j++) {
            scanf("%d", &mat[i][j]);
```

```

    }
}
for (i = 0; i < rows; i++) {
    for (j = i + 1; j < columns; j++) {
        int temp = mat[i][j];
        mat[i][j] = mat[j][i];
        mat[j][i] = temp;
    }
}
printf("\nThe transpose matrix is:\n");
for (i = 0; i < columns; i++) {
    for (j = 0; j < rows; j++) {
        printf("%d ", mat[i][j]);
        if (j == rows - 1) {
            printf("\n");
        }
    }
}
return 0;
}

```

Output:

```
Enter the number of rows and columns for the matrix: 3
3
Enter input for matrix:
1
2
3
4
5
6
7
8
9

The transpose matrix is:
1 4 7
2 5 8
3 6 9

...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise –5

Code:

```
#include<stdio.h>

int main()
{
    int mat[10][10],matn[10][10],rows,columns,i,j;
    printf("Enter the number of rows and columns for matrix:");
    scanf("%d%d",&rows,&columns);
    printf("Enter input for matrix=\n");
    for (i = 0; i < rows; i++)
    {
```

```
for (j = 0; j < columns; j++)
{
scanf("%d", &mat[i][j]);
}
}
printf("Diagonal of the Matrix=\n");
for(i=0; i<rows; i++)
{
for(j=0; j<columns; j++)
{
if(i == j)
{
printf("%d ", mat[i][j]);
}
else{
printf("\t");
}
}
printf("\n");
}
}
```

Output:

```
Enter the number of rows and columns for matrix:3
3
Enter input for matrix=
1
2
3
4
5
6
7
8
9
Diagonal of the Matrix=
1
    5
    9

...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise –6

Code:

```
#include <stdio.h>

int main() {
    int mat[10][10], rows, columns, i, j, isZeroMatrix = 1;

    printf("Enter the number of rows and columns for the matrix: ");
    scanf("%d %d", &rows, &columns);

    printf("Enter input for matrix:\n");
    for (i = 0; i < rows; i++) {
```

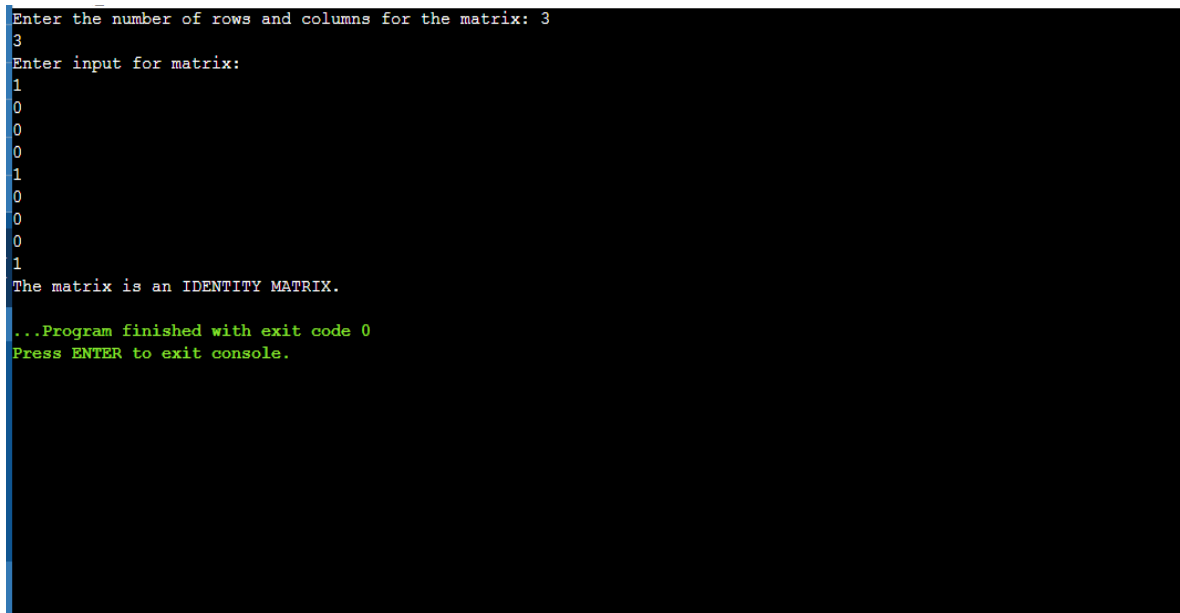
```
    for (j = 0; j < columns; j++) {  
        scanf("%d", &mat[i][j]);  
    }  
}
```

```
for (i = 0; i < rows; i++) {  
    for (j = 0; j < columns; j++) {  
        if (i == j && mat[i][j] != 1) {  
            isZeroMatrix = 0;  
            break;  
        } else if (i != j && mat[i][j] != 0) {  
            isZeroMatrix = 0;  
            break;  
        }  
    }  
}  
if (isZeroMatrix == 0) {  
    break;  
}  
}
```

```
if (isZeroMatrix == 1) {  
    printf("The matrix is an IDENTITY MATRIX.");  
} else {
```

```
        printf("The matrix is not an IDENTITY MATRIX.");  
    }  
  
    return 0;  
}
```

Output:



```
Enter the number of rows and columns for the matrix: 3  
3  
Enter input for matrix:  
1  
0  
0  
0  
1  
0  
0  
0  
1  
The matrix is an IDENTITY MATRIX.  
...Program finished with exit code 0  
Press ENTER to exit console.
```